

Algebra 1
Unit 7
Lesson 1 Practice

Name _____
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For questions 1-2, consider the expression $16y^3 - 8y^2 - 20y$.

1. Write an equivalent expression that uses the greatest common monomial factor.
2. Write an equivalent expression that does not use the greatest common monomial factor.

For questions 3-6, write an equivalent expression for the given expression that uses the greatest common monomial factor.

3. $6f^5g^3 + 12f^2g - 30f^3g^3$

4. $55s^4h^2 + 121s^6h^2 + 11s^2h$

5. $80db - 20d^7b^3$

6. $81x^5y^8 - 18x^2y^3 - 90x^6y^6$

For questions 6-9, find the greatest common monomial factor of the expression.

7. $21a^3b^4 - 49ab$

8. $24qc^2 + 20q^5c + 32q^7c^2$

9. $16r^2m^8 - 64r^8m^2 + 40r^3m^3$

10. $6x^6y^5 - 21x^7y^8 - 36x^2y^4$

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Lesson 2 Practice

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For questions 1-4, consider the trinomial $x^2 - 10x + 21$.

1. How many ^+x tiles are needed for an algebra tiles diagram of the trinomial?
2. How many ^-x tiles are needed for an algebra tiles diagram of the trinomial?
3. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 10x + 21$.

4. Complete the table for $x^2 - 10x + 21$.

	x	_____
x	x^2	_____
_____	_____	21

For questions 5-8, consider the trinomial $x^2 - 4x - 12$.

5. How many ^+x tiles are needed for an algebra tiles diagram of the trinomial?
6. How many ^-x tiles are needed for an algebra tiles diagram of the trinomial?
7. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 4x - 12$.

8. Complete the table for $x^2 - 4x - 12$.

	x	_____
x	x^2	_____
_____	_____	-12

9. Complete the table for $x^2 + x - 110$.

	x	_____
x	x^2	_____
_____	_____	-110

10. Complete the table for $x^2 + 3x - 40$.

	x	_____
x	x^2	_____
_____	_____	-40

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Lesson 3 Practice

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For questions 1-5, factor each quadratic expression using the method of your choice.

1. $b^2 + 17b + 72$

2. $h^2 - 23h + 132$

3. $x^2 + 15x + 56$

4. $y^2 - 3y - 18$

5. $x^2 - 7x - 30$

For questions 6-7, consider the trinomial $f^2 - 10f + 9$.

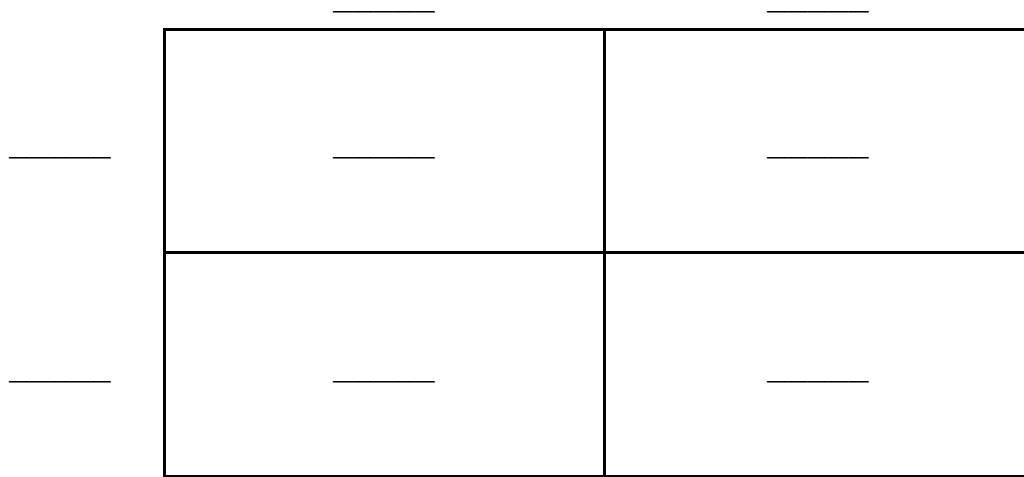
6. Complete the area model.

	f	_____
f	f^2	_____
_____	_____	9

7. What are the two binomial factors for $f^2 - 10f + 9$?

For questions 8-9, consider the trinomial $m^2 - 21m + 108$.

8. Complete the area model.



9. What are the two binomial factors for $m^2 - 21m + 108$?

10. Factor $x^2 - 12x + 35$ using factoring by grouping.

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For questions 1-5, factor each quadratic expression using the method of your choice.

1. $5b^2 + 46b + 48$

2. $4a^2 + 27a + 18$

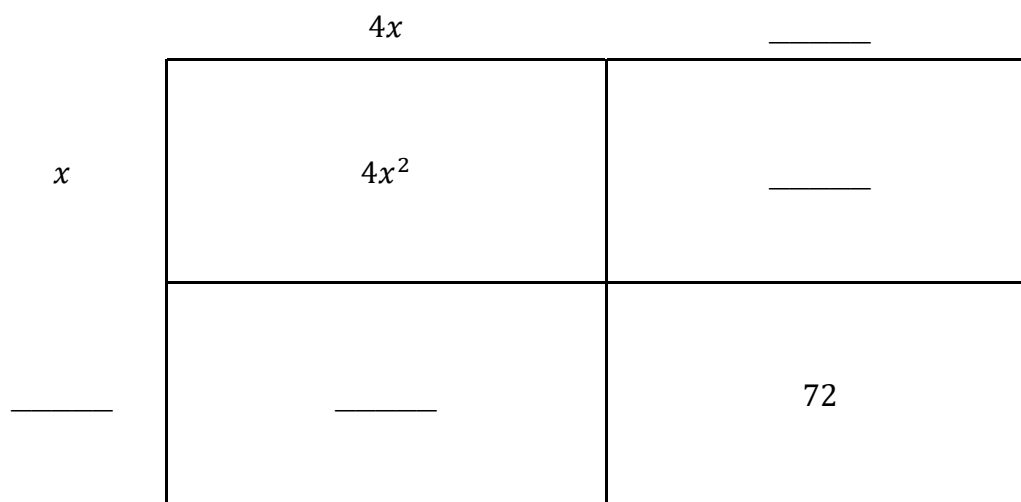
3. $2x^2 + 3x - 27$

4. $3k^2 + 5k - 28$

5. $5z^2 - 28z - 96$

For questions 6-7, consider the trinomial $4x^2 - 41x + 72$.

6. Complete the area model.



7. What are the two binomial factors for $4x^2 - 41x + 72$?

For questions 8-9, consider the trinomial $2m^2 - 9m - 110$.

8. Complete the area model.

		_____		_____
_____			_____	
_____		_____		_____

9. What are the two binomial factors for $2m^2 - 9m - 110$?

10. Factor $3x^2 - 4x - 32$ using factoring by grouping.

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Lesson 5 Practice

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For questions 1-3, factor each quadratic expression using the method of your choice.

1. $10b^2 - 9b - 9$

2. $12x^2 + 19x - 21$

3. $6d^2 - 13d + 6$

For questions 4-5, consider the trinomial $10x^2 - 61x + 72$.

4. Write an expanded form of the trinomial that can be used to factor the expression by grouping.

5. Use the expanded form to factor the expression by grouping.

For questions 6-7, consider the trinomial $8x^2 + 46x + 45$.

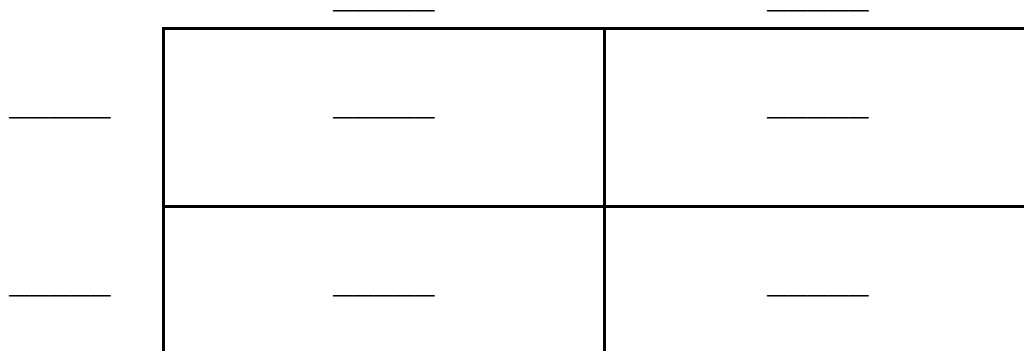
6. Complete the area model.

	4x		_____
2x	$8x^2$		_____
	_____		45

7. What are the two binomial factors for $8x^2 + 46x + 45$?

For questions 8-9, consider the trinomial $10t^2 + 9t - 9$.

8. Complete the area model.



9. What are the two binomial factors for $10t^2 + 9t - 9$?

10. Factor $6x^2 + 57x + 132$ using factoring by grouping.

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Lesson 6 Practice

Name_____

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For questions 1-6, factor each expression by grouping.

1. $2y^4 - 8y^3 + 10y^2 - 40y$

2. $15x^2 + 62x + 63$

3. $-12a + 20ab - 35b + 21$

4. $6x^2 - x - 77$

5. $3x^2 + 12x + 6x + 24$

6. $4x^2 + 38x + 18$

For questions 7-10, consider the trinomial $15x^2 - 34x - 16$.

- List all the factor pairs for -240 .
- Find the factor pair for -240 that add to the sum of -34 .
- Use question 8 to write the expanded form of $15x^2 - 34x - 16$ that can be used to factor the expression by grouping.
- Factor the expression $15x^2 - 34x - 16$ by grouping.

For questions 1-2, consider the perfect square trinomial $25x^2 + 30x + 9$.

1. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.

2. Factor the perfect square trinomial.

For questions 3-4, consider the perfect square trinomial $9x^2 - 42x + 49$.

3. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.

4. Factor the perfect square trinomial.

For questions 5-8, determine whether the quadratic expression is a perfect square trinomial. Then, factor the expression.

5. $w^2 - 24w + 144$

6. $p^2 + 2pq + q^2$

7. $49v^2 - 25$

8. $16y^2 - 256y + 32$

For questions 9-10, factor the trinomial.

9. $36a^2 - 60a + 25$

10. $p^2 - 18p + 81$

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Lesson 8 Practice

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For questions 1-2, consider the expression $36x^2 - 64y^2$.

1. Rewrite the expression as a difference of two squares.

2. Factor the expression.

For questions 3-4, consider the expression $121m^2 - 25n^2$.

3. Rewrite the expression as a difference of two squares.

4. Factor the expression.

For questions 5-8, determine whether the quadratic expression is a difference of two squares. Then, factor the expression.

5. $49s^2 - 144m^2$

6. $49g^2 + 84g + 36$

7. $9 - 4r^2$

8. $-100 + w^2$

For questions 9-10, factor the expression.

9. $x^2 - 4x + 4$

10. $81v^2 - 9p^2$